

Chapter 29: Projection Operators

Convex Analysis and Monotone Operator Theory

Based on Bauschke & Combettes (2017)

Pages 951–990

Outline

- 1 Introduction & Motivation
- 2 Basic Properties (Section 29.1)
- 3 Affine Subspaces (Section 29.2)
- 4 Projections onto Special Polyhedra (Sect. 29.3–29.4)
- 5 Epigraphs and Level Sets (Section 29.5)
- 6 Subgradient Projection Algorithms (Section 29.6)
- 7 Python Implementation Examples
- 8 Applications and Extensions
- 9 Summary & Key Takeaways

What Are Projection Operators?

- **Definition:** For a nonempty closed convex set $C \subseteq \mathcal{H}$, the *projection* of $x \in \mathcal{H}$ onto C is:

$$P_C(x) = \arg \min_{y \in C} \|x - y\|$$

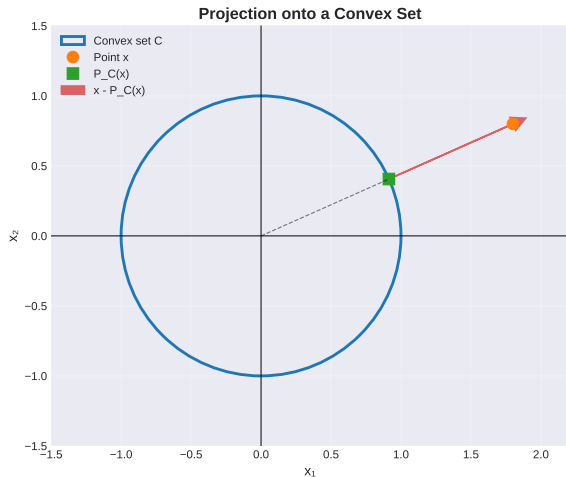
- **Key Properties:**

- 1 Uniqueness: The minimizer is unique for convex sets
- 2 Characterization: $P_C(x) \in C$ and $\langle x - P_C(x) \mid y - P_C(x) \rangle \leq 0$ for all $y \in C$
- 3 Nonexpansiveness: $\|P_C(x) - P_C(y)\| \leq \|x - y\|$

- **Applications:**

- 1 Convex optimization via projection algorithms
- 2 Constraint satisfaction in feasibility problems
- 3 Signal processing and image reconstruction
- 4 Machine learning (proximal methods)

Geometric Intuition



- The projection $P_C(x)$ is the point in C **closest** to x
- The residual vector $x - P_C(x)$ is **orthogonal** to C at $P_C(x)$
- This is the *normal cone* property
- Generalizes familiar concept: projection onto a line or plane

Key Propositions: Part I I

Proposition 29.1 (Existence and Uniqueness):

For K nonempty closed convex subset of $\mathcal{H}$ and $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$, if $C = \mathcal{R}(L)$ is the range, then:

$$P_C x = x + L^*(\gamma \text{Id} - LL^*)^{-1}(Lx)$$

for suitable $\gamma > 0$.

Proposition 29.2 (Scaled Operators):

For invertible L :

$$P_D x = L^{-1} P_C(Lx)$$

where $D = L^{-1}(C)$.

Proposition 29.3 (Product Spaces):

For finite family $(H_i)_{i \in I}$ and closed convex sets $C_i \subseteq H_i$:

$$P_{\bigoplus_i C_i} x = (P_{C_i} x_i)_{i \in I}$$

Key Propositions: Part I II

Proposition 29.5 (Nested Projections):

If $P_C x \in D$ for nonempty closed convex sets C, D :

$$P_{C \cap D} x = P_C x$$

Proposition 29.6 (Direct Sum Decomposition):

For $C \perp D$ (orthogonal decomposition):

$$P_{C+D} = P_C + P_D$$

Proposition 29.7 & 29.8:

Monotonic sequences of sets preserve properties:

- If $C_n \subseteq C_{n+1}$ and $C = \bigcup_n C_n$, then $P_C x \leftarrow P_{C_n} x$
- If $C_n \supseteq C_{n+1}$ with $\bigcap_n C_n \neq \emptyset$, then $P_C x \leftarrow P_{C_n} x$

Characterization via Variational Inequality

Core Property (Equation 29.1):

The projection $P_C(x)$ is characterized by:

$$\forall y \in C : \quad \langle x - P_C(x) \mid y - P_C(x) \rangle \leq 0$$

Interpretation: The residual vector $x - P_C(x)$ makes an **obtuse angle** with any direction $y - P_C(x)$ pointing into C .

Equivalence: This is equivalent to saying $x - P_C(x) \in N_C(P_C(x))$, where N_C is the *normal cone* to C at $P_C(x)$.

Lemma 29.12:

For $\alpha, \beta \in \mathbb{R}_{++}$ with $\alpha < \beta$ and $D = -\gamma + C$ where $\gamma = \beta/\alpha$:

$$\phi(\beta) = \beta^{-1} \|P_D(\beta x)\|$$

is decreasing in β .

Nonexpansiveness and Firm Nonexpansiveness

Corollary 29.12 (Key Inequality):

Define $\phi(\alpha) = \alpha^{-1} \|P_C(\alpha x) - y\|$ for $0 < \alpha < \beta$ and any $y \in \mathcal{H}$.

Then ϕ is **decreasing in** α .

Proposition 29.11 (Lipschitz Continuity):

For C nonempty closed convex and $\gamma \in]1, +\infty[$:

$$\|P_C(\gamma x)\| \leq \gamma \|P_C x\|$$

Firm Nonexpansiveness:

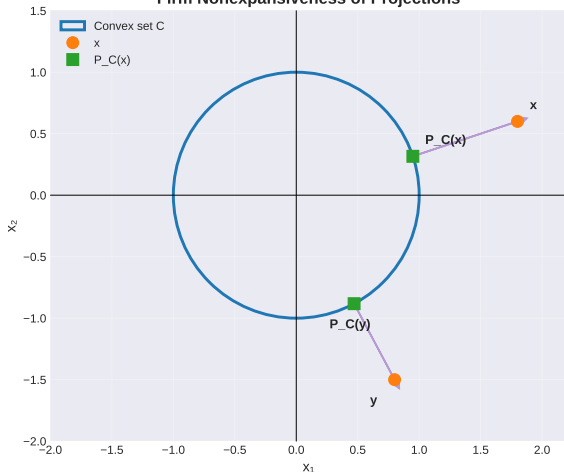
The projection operator satisfies:

$$\|P_C x - P_C y\|^2 \leq \langle x - y \mid P_C x - P_C y \rangle$$

This is *stronger* than ordinary nonexpansiveness!

Visualizing Nonexpansiveness

Firm Nonexpansiveness of Projections



- Projection operators *contract* distances between points
- The contraction is *firm*: stronger than simple nonexpansiveness
- Distance between projections $\leq$ distance between original points
- This property is **critical** for convergence analysis

Projections onto Affine Subspaces I

Proposition 29.14:

For closed affine subspace C and $x, y \in \mathcal{H}$, $z \in C$:

- (i) P_C is weakly continuous
- (ii) $x - P_C x \perp C - C$ (orthogonal to direction space)
- (iii) $\|x - P_C y\|^2 = \|P_C x - P_C y\|^2 + \|(\text{Id} - P_C)x - (\text{Id} - P_C)y\|^2$
- (iv) $\|(2P_C - \text{Id})x - (2P_C - \text{Id})y\| = \|x - y\|$ (reflection)
- (v) $\|P_C x - P_C y\|^2 = \langle x - y \mid P_C x - P_C y \rangle$
- (vi) $\|(\text{Id} - P_C)x - (\text{Id} - P_C)y\|^2 = \langle x - y \mid (\text{Id} - P_C)x - (\text{Id} - P_C)y \rangle$
- (vii) $\|x - P_C x\|^2 = \langle x - z \mid x - P_C x \rangle$

Projections onto Affine Subspaces II

Proposition 29.15 (Orthonormal Basis):

If $(e_i)_{i \in I}$ is orthonormal with $C = \overline{\text{span}}\{e_i\}$:

$$P_C x = \sum_{i \in I} \langle x | e_i \rangle e_i$$

Example: Projection onto line $U = \mathbb{R}u$ with $\|u\| = 1$:

$$P_U x = \langle x | u \rangle u$$

Proposition 29.16:

For weighted Hilbert space with weights $(w_i)_{i \in I}$:

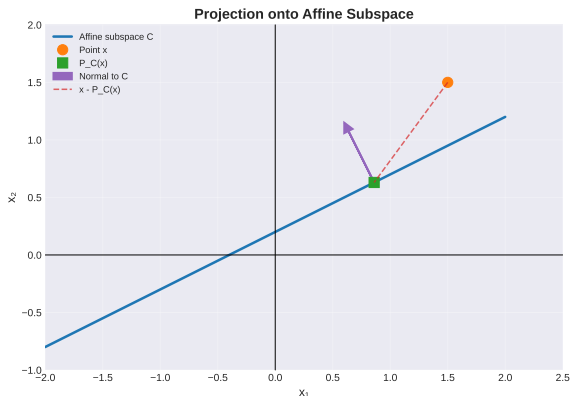
$$\langle x, y \rangle = \sum_{i \in I} w_i \langle x_i | y_i \rangle$$

Projections onto Affine Subspaces III

The projection onto $D = \{(x)_{i \in I} : x \in \mathcal{H}\}$ is:

$$P_D x = (p)_{i \in I} \text{ where } p = \sum_{i \in I} w_i x_i$$

Geometric Illustration



- Affine subspaces are **translates** of linear subspaces
- Projection is **orthogonal** to the subspace direction
- For orthonormal bases, projection is **simply** the sum of coordinate components
- Used extensively in least-squares problems

Projections onto Half-Spaces I

Example 29.20 (Half-Space):

For $u \in \mathcal{H} \setminus \{0\}$ and $\eta \in \mathbb{R}$, the half-space:

$$C = \{x \in \mathcal{H} : \langle x | u \rangle \leq \eta\}$$

Exactly one of the following holds:

- (i) $u = 0$ and $\eta \geq 0$: then $C = \mathcal{H}$ and $P_C = \text{Id}$
- (ii) $u = 0$ and $\eta < 0$: then $C = \emptyset$
- (iii) $u \neq 0$:

$$P_C x = \begin{cases} x & \text{if } \langle x | u \rangle \leq \eta \\ x + \frac{\eta - \langle x | u \rangle}{\|u\|^2} u & \text{if } \langle x | u \rangle > \eta \end{cases}$$

Projections onto Half-Spaces II

Example 29.21 (Slab):

For $\eta_1 \leq \eta_2$ and nonzero u :

$$C = \{x \in \mathcal{H} : \eta_1 \leq \langle x | u \rangle \leq \eta_2\}$$

With $\phi : t \mapsto \sum_{i \in I} \max\{\xi_i, t, 0\}$, the projection is:

$$P_C x = \begin{cases} x + \frac{\eta_1 - \langle x | u \rangle}{\|u\|^2} u & \text{if } \langle x | u \rangle < \eta_1 \\ x & \text{if } \eta_1 \leq \langle x | u \rangle \leq \eta_2 \\ x + \frac{\eta_2 - \langle x | u \rangle}{\|u\|^2} u & \text{if } \langle x | u \rangle > \eta_2 \end{cases}$$

Projections onto Half-Spaces III

Proposition 29.22 (Two Parallel Half-Spaces):

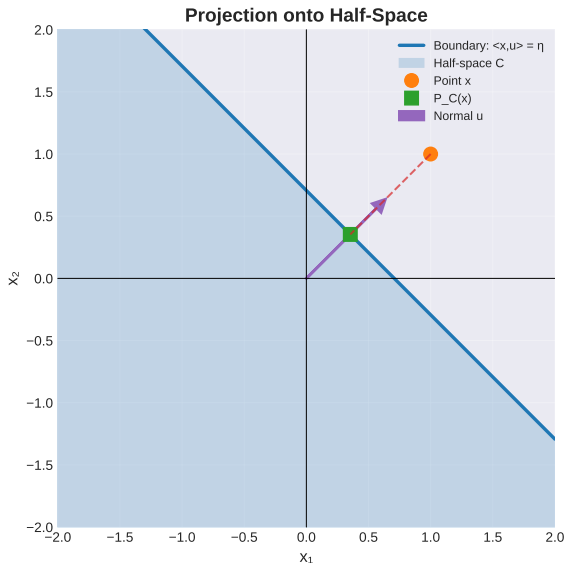
For linearly independent u_1, u_2 with $\eta_1, \eta_2 \in \mathbb{R}$:

$$C = \{x : \langle x | u_1 \rangle \leq \eta_1\} \cap \{x : \langle x | u_2 \rangle \leq \eta_2\}$$

Closed-form expressions for $P_C x$ in terms of:

$$\nu_1 = \frac{\|u_2\|^2(\langle x | u_1 \rangle - \eta_1) - \langle u_1 | u_2 \rangle(\langle x | u_2 \rangle - \eta_2)}{\|u_1\|^2\|u_2\|^2 - \langle u_1 | u_2 \rangle^2}$$

Half-Space Projection Example



- Simplest nontrivial case: *half-space*
- Projection direction is **normal** to the boundary
- If x already satisfies constraint, no movement
- Otherwise, move along normal by amount:

$$\frac{\eta - \langle x | u \rangle}{\|u\|^2}$$

Projections onto Simplices I

Proposition 29.26 (Projection onto Convex Hull):

For finite set $\{x_1, \dots, x_m\}$ and $C = \text{conv}\{x_i\}_{i \in I}$:

$$P_C x = \sum_{i \in I} \bar{\alpha}_i x_i$$

where $(\bar{\alpha}_i)$ solves:

$$\min_{\alpha \in \mathbb{R}_+^m, \sum \alpha_i = 1} \sum_{i,j} \alpha_i \alpha_j \langle x_i | x_j \rangle - 2 \sum_i \alpha_i \langle x_i | x \rangle$$

Example 29.27 (Probability Simplex):

For $\mathcal{H} = \mathbb{R}^N$ and:

$$\Delta = \{x \in \mathbb{R}_+^N : \sum_i x_i = 1\}$$

Projections onto Simplices II

The projection P_{Δ} can be computed via:

$$P_{\Delta} = \text{Id} - \text{Prox}_f$$

where f is determined by the simplex structure.

Projections onto Simplices III

Example 29.28 (ℓ^1 Unit Ball):

For $C = \{x \in \mathcal{H} : \|x\|_1 \leq 1\}$:

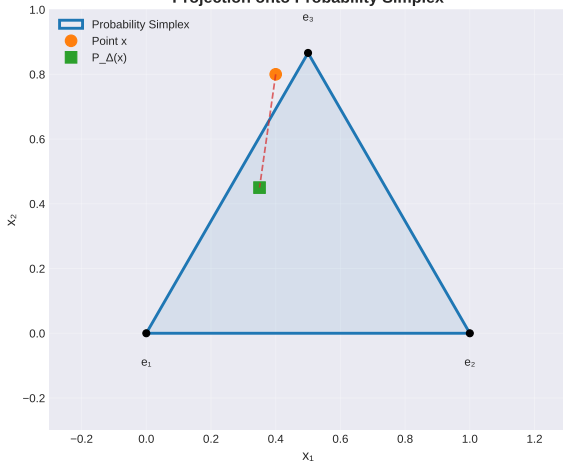
$$P_C x = (\text{sign}(\xi_i) \pi_i)_{i \in I}$$

where $\pi = P_\Delta(|\xi|)$ and Δ is the probability simplex.

Uses coordinate-wise thresholding combined with probability simplex projection.

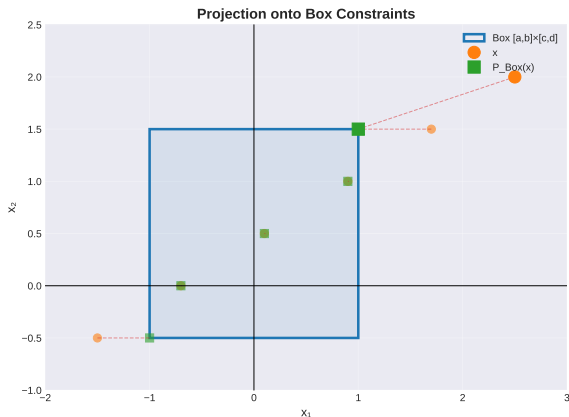
Simplex Projection Visualization

Projection onto Probability Simplex



- **Probability simplex:** points with non-negative coordinates summing to 1
- **Important in:** machine learning, statistics, economics
- Projection can be solved *efficiently* using sorting
- Soft-thresholding operations are key ingredient
- Example: sparse recovery, compressed sensing

Box Constraint Projection



- **Box constraints:** $C = [a, b] \times [c, d] \times \dots$
- **Element-wise clipping:**

$$P_C(x) = (\text{clip}(x_1, a, b), \text{clip}(x_2, c, d), \dots)$$

where $\text{clip}(\xi, a, b) = \max(a, \min(\xi, b))$

- **Complexity:** $O(n)$ for n dimensions
- **Application:** bounded optimization, imaging

Projections onto Epigraphs I

Proposition 29.35:

Let $f : \mathcal{H} \rightarrow \mathbb{R}$ be convex continuous, $C = \text{epi } f$, and $(z, \zeta) \in \mathcal{H} \times \mathbb{R}$. Then the inclusion $z \in x + (f(x) - \zeta)\partial f(x)$ has unique solution $\bar{x}$, and:

$$P_C(z, \zeta) = (\bar{x}, f(\bar{x}))$$

Example 29.36:

For quadratic $f(x) = \frac{1}{2} \sum_i \alpha_i \xi_i^2$, projection yields:

$$\bar{\xi}_i = \frac{\zeta_i}{(\eta - \zeta)\alpha_i + 1}$$

where $\eta = f(\bar{x})$ solves the one-dimensional equation.

Projections onto Epigraphs II

Proposition 29.37:

For $f \in \Gamma_0(\mathcal{H})$ with $\zeta \in \text{int } f(\mathcal{H})$, $f(\zeta)$ and $C \subseteq \text{dom } f$:

The equation $f(\text{Prox}_{\partial f} z) = \zeta$ has at least one solution.

If $\bar{\nu}$ is a solution, then:

$$P_C z = \text{Prox}_{\partial f} \bar{\nu} z$$

Projections onto Lower Level Sets

Proposition 29.35 (Continued):

For epigraph $C = \text{epi } f$ where f is convex continuous:

The inclusion $z \in x + (f(x) - \zeta)\partial f(x)$ provides the projection characterization.

Key Insight:

Projection onto level sets $\text{lev}_{\leq \zeta} f = \{x : f(x) \leq \zeta\}$ reduces to:

- 1 If $f(x) \leq \zeta$: $P_C x = x$ (already feasible)
- 2 If $f(x) > \zeta$: Move along subdifferential direction $\partial f(x)$ until reaching level ζ

Application:

Critical for *level set projection algorithms* used in constrained optimization.

Subgradient Projector and Fixed Points I

Motivation:

For projection operator onto epi-graph, define:

$$Gx = \begin{cases} P_C(x + \alpha_n \frac{\mu - f(x_n)}{s(x_n)} s(x_n)) & \text{if } s(x_n) \neq 0 \\ x_n & \text{if } s(x_n) = 0 \end{cases}$$

where s is a selection of ∂f .

Key Examples:

Example 29.47:

For $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \max\{x + 1, 2x + 1\}$:

$$Gx = \begin{cases} x & \text{if } x \leq -1 \\ -1 & \text{if } -1 < x < 0 \\ [-1/2, -1/2] & \text{if } x = 0 \\ -1/2 & \text{if } x > 0 \end{cases}$$

Subgradient Projector and Fixed Points II

So G is **discontinuous** despite f being smooth everywhere!

Subgradient Projector and Fixed Points III

Example 29.48

For $\mathcal{H} = \ell^2(\mathbb{N} \setminus \{0\})$ with:

$$f : (\zeta_n) \mapsto \sum_{n \in \mathbb{N}} n \zeta_n^{2n}$$

The subgradient projector: - Is **not weakly continuous** - Is **not nonexpansive** - Despite f being convex and continuous

These examples show limitations of subgradient approaches!

Polyak's Subgradient Projection Algorithm I

Corollary 29.50 (Polyak's Algorithm):

Let C be closed convex, $f : \mathcal{H} \rightarrow \mathbb{R}$ continuous convex with $\text{Argmin}_C f \neq \emptyset$.

Suppose one of:

- (i) f bounded on every bounded subset
- (ii) f^* supercoercive
- (iii) $\mathcal{H}$ finite-dimensional

Let $(\alpha_n)_{n \in \mathbb{N}}$ be sequence with $0 < \inf_n \alpha_n \leq \sup_n \alpha_n < 2$, s a selection of ∂f , $x_0 \in C$, $\mu = \min f(C)$:

Algorithm:

$$x_{n+1} = \begin{cases} P_C \left(x_n + \alpha_n \frac{\mu - f(x_n)}{s(x_n)^2} s(x_n) \right) & \text{if } s(x_n) \neq 0 \\ x_n & \text{if } s(x_n) = 0 \end{cases}$$

Polyak's Subgradient Projection Algorithm II

Convergence Result:

Then $(x_n)_{n \in \mathbb{N}}$ converges weakly to a minimizer of f over C .

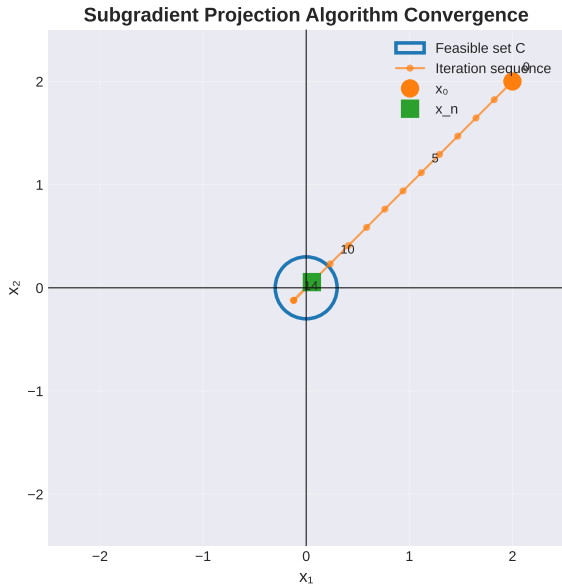
Key Advantages:

- ① Works with *only* subgradients (not full derivatives)
- ② Handles non-smooth objectives
- ③ Step size $\mu - f(x_n)$ is *adaptive*
- ④ Natural normalization by $\|s(x_n)\|^2$

Key Limitations:

- ① Requires knowledge of minimum value μ
- ② Weak convergence only (not strong)
- ③ Slow for ill-conditioned problems

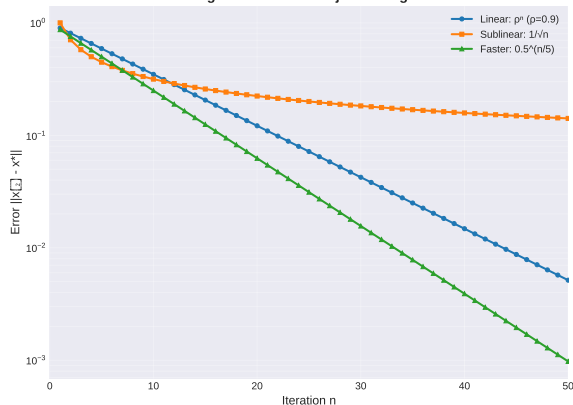
Algorithm Visualization and Convergence



- Iterates spiral inward toward feasible region
- Step size adapts based on constraint violation
- Projection ensures feasibility at each step
- Convergence guaranteed under mild conditions

Convergence Rates Comparison

Convergence Rates of Projection Algorithms



- **Linear:** $\|e_n\| \sim \rho^n$ (fastest, $\rho < 1$)
- **Sublinear:** $\|e_n\| \sim 1/\sqrt{n}$ (slow but guaranteed)
- **Behavior:**
 - Problem conditioning matters
 - Smooth problems: fast
 - Non-smooth: slower
 - Degenerate: sublinear
- **Practical:** Accelerated variants often needed

Basic Projection Implementation

```
import numpy as np
from scipy.optimize import minimize

def proj_box(x, a, b):
    """Projection onto box  $[a,b]^n$ """
    return np.clip(x, a, b)

def proj_ball(x, radius=1.0):
    """Projection onto Euclidean ball"""
    norm_x = np.linalg.norm(x)
    if norm_x <= radius:
        return x.copy()
    return (radius / norm_x) * x

def proj_halfspace(x, u, eta):
    """Projection onto  $\{y : \langle y, u \rangle \leq \eta\}$ """
    inner = np.dot(x, u)
    if inner <= eta:
        return x.copy()
    return x - (inner - eta) * u / np.dot(u, u)

# Test
x = np.array([2.0, 3.0])
print(f"Proj onto ball: {proj_ball(x, radius=2)}")
print(f"Proj onto box: {proj_box(x, a=0, b=2)}")
print(f"Proj onto HS: {proj_halfspace(x, u=np.ones(2), eta=1)}")
```

Probability Simplex Projection

```
def proj_simplex(x):  
    """Project x onto probability simplex  
    min ||y - x||^2 s.t. y >= 0, sum(y) = 1  
    """  
    n = len(x)  
    # Sort x in descending order  
    u = np.sort(x)[::-1]  
  
    # Find rho: largest i such that u[i] > (sum(u[:i+1]) - 1) / (i+1)  
    cssv = np.cumsum(u)  
    rho = np.where(u > (cssv - 1) / np.arange(1, n+1))[0][-1]  
  
    # Compute threshold  
    theta = (cssv[rho] - 1) / (rho + 1)  
  
    # Apply threshold and rectify  
    y = np.maximum(x - theta, 0)  
    return y  
  
# Example: project outside simplex  
x = np.array([2.0, 0.5, -1.0])  
y = proj_simplex(x)  
print(f"Original: {x}")  
print(f"Projected: {y}")  
print(f"Valid: {y.sum()}, nonneg={np.all(y >= 0)}")
```

Alternating Projections Algorithm

```
def alternating_projections(P1, P2, x0, max_iter=1000, tol=1e-6):
```

```
    """
```

```
    Alternating projections:  $x_{n+1} = P2(P1(x_n))$ 
```

```
    Used for finding point in intersection of two sets
```

```
    """
```

```
    x = x0.copy()
```

```
    history = [np.linalg.norm(x)]
```

```
    for n in range(max_iter):
```

```
        x_prev = x.copy()
```

```
        x = P2(P1(x))    # Alternate between projections
```

```
        error = np.linalg.norm(x - x_prev)
```

```
        history.append(error)
```

```
        if error < tol:
```

```
            print(f"Converged in {n+1} iterations")
```

```
            return x, history
```

```
    print(f"Max iterations reached")
```

```
    return x, history
```

```
# Example: intersection of two balls
```

```
def proj_ball1(x):
```

```
    return proj_ball(x - np.array([0.5, 0]), radius=1.0) + np.array([0.5, 0])
```

```
def proj_ball2(x):
```

Proximal Operator and Projection Connection

```
def prox_convex_fn(f, grad_f, x, gamma=0.1, max_iter=20):
    """Compute prox_gamma_f(x) using gradient descent"""
    y = x.copy()
    for _ in range(max_iter):
        y = y - gamma * (grad_f(y) + 2/gamma * (y - x))
    return y

def proj_epi(f, grad_f, x, eta):
    """Project onto epigraph of f: epi f = {(x,t) : f(x) <= t}"""
    # If already in epigraph, stay put
    if f(x) <= eta:
        return x, eta

    # Otherwise, minimize distance to epigraph
    # This involves solving a subdifferential inclusion
    x_proj = prox_convex_fn(f, grad_f, x)
    f_proj = f(x_proj)

    if f_proj < eta:
        # Project onto boundary
        return x_proj, f_proj
    else:
        # On the boundary
        return x_proj, eta

# Example: f(x) = ||x||^2
f = lambda x: np.sum(x**2)
```

Key Applications

1 Convex Feasibility

- Find point in $C_1 \cap C_2 \cap \dots \cap C_m$
- Use alternating projections: $x_{n+1} = P_{C_{i_n}}(x_n)$
- Guarantees weak convergence under standard conditions

2 Constrained Optimization

- Minimize $f(x)$ subject to $x \in C$
- Combine gradient/subgradient with projection
- Example: Projected gradient descent $x_{n+1} = P_C(x_n - \gamma \nabla f(x_n))$

3 Image Processing

- Denoise via projection onto convex set of “good” images
- Convex constraints: total variation, sparsity, etc.
- Fast iterative methods essential

4 Signal Recovery

- Compressed sensing: recover sparse signal from few measurements
- Projection onto ℓ_1 ball or simplex
- Efficient algorithms critical

Connection to Proximal Methods

Proximal Operator and Projection:

The *proximal operator* is a generalization of projections:

$$\text{prox}_{\gamma f}(x) = \arg \min_y \left\{ f(y) + \frac{1}{2\gamma} \|y - x\|^2 \right\}$$

When $f = \iota_C$ (indicator function of C):

$$\text{prox}_{\gamma \iota_C}(x) = P_C(x)$$

Splitting Methods:

For $f = g + h$:

$$\text{Forward-Backward Splitting : } x_{n+1} = \text{prox}_{\gamma h}(x_n - \gamma \nabla g(x_n))$$

When $g = 0$, reduces to fixed point of proximal operator.

When $h = \iota_C$, becomes projected gradient method!

Main Results Summary

Uniqueness Projections onto nonempty closed convex sets are unique and well-defined.

Characterization Via variational inequality: $\langle x - P_C(x) \mid y - P_C(x) \rangle \leq 0$ for all $y \in C$.

Firmness Projections are *firmly nonexpansive* operators (stronger property).

Explicit Cases Explicit formulas for half-spaces, simplices, ellipsoids, boxes, etc.

Convergence Subgradient projection methods converge weakly to constrained minimizers.

Complexity Many projections have efficient algorithms ($O(n)$ or $O(n \log n)$).

Generalizations Proximal operators generalize projections; splitting methods unify many algorithms.

Important Insights

- **Projection is orthogonal:** The residual $x - P_C(x)$ is perpendicular to all directions into C .
- **Contraction property:** Projections reduce distances between points (firm nonexpansiveness).
- **Versatility:** Same operation (projection) used in optimization, feasibility, and fixed-point problems.
- **Structure matters:** Different set structures (half-space, simplex, box) have different computational costs.
- **Weak convergence:** Most projection algorithms converge weakly; strong convergence requires additional structure.
- **Generalization:** Proximal operators and splitting methods extend projection ideas to broader optimization problems.

Further Reading

- **Bauschke & Combettes (2017):** “Convex Analysis and Monotone Operator Theory in Hilbert Spaces”
 - Chapters 24–30 cover projections, proximal methods, and splitting algorithms
 - Comprehensive treatment with many examples
- **Boyd & Parikh (2011):** “Proximal Algorithms”
 - Practical implementation guidance
 - Applications to machine learning and statistics
- **Nesterov (2018):** “Lectures on Convex Optimization”
 - Convergence analysis with complexity bounds
 - Accelerated methods
- **Recent Applications:**
 - Machine learning: federated learning, distributed optimization
 - Signal processing: compressed sensing, sparse recovery
 - Computer vision: image denoising, inpainting